

SensiSpace: A Biologically-Inspired Deep Learning Architecture for Satellite Land Cover Classification

Yash Lohia

Department of Computer Science and Engineering

ISRO RESPOND Initiative - Major Project

Abstract

Automated land cover classification from satellite imagery is critical for environmental monitoring, urban planning, and resource management. This paper presents SensiSpace, a satellite image classification system featuring a novel biologically-inspired architecture called Baby Dragon Hatchling (BDH). BDH incorporates four neuroscience-inspired modules: (1) Multi-Scale Feature Pyramids for capturing objects at varying spatial scales, (2) Lateral Inhibition Blocks that sharpen feature boundaries by mimicking retinal neural circuits, (3) Contextual Attention combining channel and spatial attention for selective feature focusing, and (4) Recurrent Refinement that iteratively improves feature representations through gated feedback loops. Evaluated on the EuroSAT benchmark dataset (27,000 Sentinel-2 satellite images across 10 land cover classes), BDH achieves 95.87% classification accuracy with approximately 2.03M parameters, outperforming a transfer-learning-based ResNet-18 baseline (81.98% accuracy) by 13.89 percentage points. We further integrate a spectral color-space segmentation module for pixel-level land cover visualization. The complete system is deployed as a full-stack web application using FastAPI and PyTorch.

Keywords: Satellite Remote Sensing, Land Cover Classification, Biologically-Inspired Neural Networks, Convolutional Neural Networks, EuroSAT, Attention Mechanisms

I. INTRODUCTION

Satellite remote sensing generates massive volumes of high-resolution imagery used by organizations such as ISRO (Indian Space Research Organisation) and ESA (European Space Agency) for Earth observation. Manual analysis of this data is impractical at scale, motivating the need for automated classification systems.

Conventional CNN architectures like ResNet, while effective for natural image classification via transfer learning, are not purpose-built for the unique characteristics of satellite imagery -- namely multi-scale land cover patterns, subtle spectral boundaries between adjacent land types, and the need for contextual understanding of geospatial features.

Contributions of this paper:

- Baby Dragon Hatchling (BDH): A novel biologically-inspired CNN incorporating lateral inhibition, dual attention, multi-scale extraction, and recurrent refinement.
- Comprehensive benchmarking against ResNet-18 with ImageNet transfer learning on EuroSAT under identical training conditions.
- Spectral segmentation module for pixel-level land cover visualization using HSV/LAB color-space analysis.
- Full-stack web deployment for real-time satellite image classification and visualization.

II. RELATED WORK

A. CNNs for Remote Sensing

Helber et al. [1] introduced the EuroSAT dataset and demonstrated that CNNs like ResNet-50 and GoogLeNet achieve strong performance on Sentinel-2 imagery. Transfer learning from ImageNet has been widely adopted [2], though domain mismatch between

natural and satellite images can limit effectiveness.

B. Attention Mechanisms

Squeeze-and-Excitation Networks [3] introduced channel attention. CBAM [4] extended this with spatial attention. These mechanisms enable selective focus on informative features, particularly relevant for satellite imagery where different spectral channels carry varying importance.

C. Multi-Scale Feature Extraction

Inception modules [5] demonstrated effectiveness of parallel convolutions with different kernel sizes. Feature Pyramid Networks [6] showed that multi-scale representations are critical for detecting objects at different scales.

D. Biologically-Inspired Models

Lateral inhibition, first described by Hartline et al. [7] in the horseshoe crab retina, is fundamental in biological visual processing. Recurrent processing in the visual cortex [8] has inspired feedback mechanisms, but these remain underexplored for geospatial applications.

III. DATASET - EUROSAT

The EuroSAT dataset [1] consists of 27,000 georeferenced satellite images captured by the Sentinel-2 satellite (European Space Agency). Each image is 64x64 pixels in RGB with 10m spatial resolution.

A. Dataset Properties

Property	Value
Source	Sentinel-2 (ESA)
Total Images	27,000
Resolution	64x64 RGB
Classes	10
Train/Test Split	80% / 20%
Spatial Resolution	10 m/pixel

B. Land Cover Classes

Class	Description
AnnualCrop	Wheat, corn, rice fields
Forest	Dense deciduous/coniferous tree cover
HerbaceousVegetation	Grasslands, meadows, scrubland
Highway	Major roads and expressways
Industrial	Factories, warehouses
Pasture	Grazing land for livestock
PermanentCrop	Orchards, vineyards
Residential	Urban residential areas
River	Rivers, streams, canals
SeaLake	Seas, lakes, reservoirs

IV. PROPOSED ARCHITECTURE - BABY DRAGON HATCHLING

A. Architecture Overview

BDH is a three-stage deep learning architecture where each stage chains three biologically-inspired modules -- Multi-Scale Feature Pyramid, Lateral Inhibition Block, and Contextual Attention -- followed by a Recurrent Refinement module before classification.

```
Input (3x64x64)
-> Stem: Conv3x3->BN->ReLU->Conv3x3->BN->ReLU [3->64 ch]
-> BDH Stage 1 (64 ch) -> MaxPool 2x2 -> 32x32
-> BDH Stage 2 (128 ch) -> MaxPool 2x2 -> 16x16
-> BDH Stage 3 (256 ch) -> MaxPool 2x2 -> 8x8
-> Recurrent Refinement (2 iterations)
-> Global Avg Pool (256x1x1)
-> Dropout(0.3)->FC(256->128)->ReLU->Dropout(0.2)->FC(128->10)
```

Each BDH Stage:

```
Input -> MultiScale -> LateralInhibition -> ContextualAttention -> ADD -> ReLU
|                                     ^
+----- 1x1 Conv Residual Projection -----+
```

B. Module 1: Multi-Scale Feature Pyramid

Satellite images contain objects at vastly different spatial scales. BDH processes input through three parallel convolution branches with kernels of 3x3 (fine edges), 5x5 (medium structures), and 7x7 (large patterns). Outputs are concatenated along the channel dimension:

```
F_ms(x) = Concat[f_3x3(x), f_5x5(x), f_7x7(x)]
where f_kxk(x) = ReLU(BN(Conv_kxk(x)))
```

C. Module 2: Lateral Inhibition Block

Inspired by retinal neural circuits where neighboring neurons suppress each other to sharpen edges. A depthwise convolution estimates surround activity, which is subtracted with learnable strength α (initialized to 0.3):

```
F_li(x) = ReLU(x - alpha * BN(DWConv_3x3(x)))
alpha is a learnable nn.Parameter
```

This sharpens feature boundaries -- critical for distinguishing adjacent land types (e.g., where a river meets a forest).

D. Module 3: Contextual Attention

A dual attention mechanism combining channel and spatial attention:

Channel Attention (WHAT): GAP -> FC(C->C/8) -> ReLU -> FC(C/8->C) -> Sigmoid. Produces per-channel weights (SE-Net style, reduction $r=8$).

Spatial Attention (WHERE): AvgPool + MaxPool across channels -> Concat -> Conv7x7 -> Sigmoid. Produces per-pixel attention map.

Input is sequentially reweighted: first by channel importance, then by spatial importance.

E. Module 4: Recurrent Refinement

Mimics the brain's top-down feedback pathways. Over 2 iterations, features are iteratively refined through a gated mechanism:

```
refined = ReLU(BN(Conv3x3(x)))
gate = Sigmoid(Conv1x1(refined))
x_new = x * gate + refined * (1 - gate)
```

The sigmoid gate decides per-element whether to keep original features (gate near 1) or replace with refined features (gate near 0).

F. Parameter Summary

Component	Parameters
Stem (2x Conv3x3 + BN)	~19K
BDH Stage 1 (64->64)	~60K

BDH Stage 2 (64->128)	~200K
BDH Stage 3 (128->256)	~750K
Recurrent Refinement	~720K
Classifier (FC)	~34K
TOTAL	~2,032,275

V. BASELINE - RESNET-18

We use ResNet-18 [9] as a CNN baseline with ImageNet transfer learning. The convolutional base is frozen, and only the final fully connected layer FC(512->10) is trainable, yielding ~5,130 trainable parameters. The total ResNet-18 architecture contains 11.2M parameters (11.17M frozen).

VI. EXPERIMENTAL SETUP

A. Training Configuration

Setting	Value
Optimizer	Adam
Learning Rate	1e-3
LR Scheduler	CosineAnnealingLR
Loss Function	CrossEntropyLoss
Batch Size	32
Epochs	10

B. Data Augmentation

Transform	Value
Resize	64x64
Random H/V Flip	p=0.5
Random Rotation	+/-15 deg
Color Jitter	brightness=0.2, contrast=0.2
Normalization	ImageNet mean/std

VII. RESULTS AND ANALYSIS

A. Performance Comparison

Metric	ResNet-18	BDH	Winner
Accuracy	81.98%	95.87%	BDH
Precision	81.64%	95.78%	BDH
Recall	81.51%	96.01%	BDH
F1-Score	81.51%	95.85%	BDH
Inference (ms)	0.18	0.60	ResNet
Train Time (s)	245.0	3658.4	ResNet
Parameters	5,130	2,032,275	ResNet

B. Analysis

Accuracy Superiority: BDH outperforms ResNet-18 by 13.89 percentage points across all metrics. The improvement is consistent across precision, recall, and F1, indicating robust performance across all 10 classes.

Parameter Efficiency: While BDH has more trainable parameters (2.03M vs 5.1K), ResNet-18's total architecture is 11.2M (11.17M frozen). BDH achieves higher accuracy with ~5.5x fewer total parameters, demonstrating domain-specific architecture beats brute-force scaling.

Inference Speed: ResNet-18 is ~3.3x faster (0.18ms vs 0.60ms), but BDH's 0.60ms still enables >1,600 images/second -- well within real-time requirements.

Training Time: BDH requires 15x more training time (3,658s vs 245s) because it trains all parameters from scratch. This is a one-time cost not affecting deployment.

C. Architectural Feature Comparison

Feature	ResNet-18	BDH
Type	Standard CNN	Bio-Inspired CNN
Multi-Scale	No (3x3 only)	Yes (3x3+5x5+7x7)
Attention	None	Channel + Spatial
Lateral Inhibition	None	Yes (learnable)
Recurrent Feedback	None	Yes (2 iterations)
Skip Connections	Additive residual	Projected residual
Pre-trained	Yes (ImageNet)	No (from scratch)
Accuracy	81.98%	95.87%

VIII. SPECTRAL LAND COVER VISUALIZATION

Standard Grad-CAM produces blurry, low-resolution activation maps. We implement spectral segmentation based on color-space analysis -- the same principle underlying NDVI in professional remote sensing.

A. Method

- Convert RGB to HSV and CIELAB color spaces
- Compute Green Leaf Index: $GLI = (2G - R - B) / (2G + R + B + 1)$ for vegetation detection
- HSV hue analysis: green -> vegetation, blue -> water, low saturation -> urban
- LAB a-channel for urban warmth, b-channel for bare soil
- Threshold into 6 masks: dense vegetation, light vegetation, water, urban, road, bare soil
- Map spectral categories to EuroSAT classes using model softmax probabilities
- Render colored overlays with contour borders and confidence labels

B. Spectral-to-Class Mapping

Spectral Type	Candidate EuroSAT Classes
Dense Vegetation	Forest, HerbaceousVegetation, Pasture
Light Vegetation	AnnualCrop, PermanentCrop, Pasture
Water	River, SeaLake
Urban	Residential, Industrial
Road	Highway
Bare Soil	AnnualCrop, PermanentCrop, Industrial

IX. SYSTEM ARCHITECTURE

A. Technology Stack

Layer	Technology
Deep Learning	PyTorch
Backend API	FastAPI (Python)
Frontend	HTML5, CSS3, JavaScript
Image Processing	OpenCV, Pillow
Metrics	scikit-learn
Server	Uvicorn ASGI

B. API Endpoints

Endpoint	Method	Description
/api/predict	POST	Both models predictions + spectral overlay
/api/sample	GET	Random EuroSAT sample image
/api/metrics	GET	Training benchmark data
/api/colors	GET	RGB color palette for 10 classes
/api/health	GET	System health check

C. Frontend

Premium dark-mode web UI with three tabbed views: (1) ResNet-18 results with probability charts and spectral overlay, (2) BDH results, (3) Head-to-head comparison with full benchmark table and side-by-side attention maps. Features glassmorphism design, micro-animations, and drag-and-drop image upload.

X. CONCLUSION

This paper presents Baby Dragon Hatchling (BDH), a biologically-inspired deep learning architecture for satellite land cover classification. BDH achieves 95.87% accuracy on EuroSAT -- a 13.89 percentage point improvement over ResNet-18 with transfer learning -- while using 5.5x fewer total architecture parameters. The four key innovations -- multi-scale feature pyramids, lateral inhibition, contextual attention, and recurrent refinement -- each contribute by mimicking how the human visual cortex processes complex spatial information.

Future Work:

- Extending BDH to multi-spectral (13-band) Sentinel-2 data beyond RGB
- Semantic segmentation variant for pixel-level classification
- Scaling to BigEarthNet (590K images)
- Lightweight model distillation for edge/satellite hardware deployment
- Integration with real-time satellite data feeds for continuous monitoring

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